SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets
Assessment Tool Weightage: 30%

QUESTIONS: 05 **Total Marks:50**

Annexure A
MOCK INTERVIEW

Criteria	Technical Knowledge *(2.5)*	Problem Solving *(2.5)*	Analytical Thinking *(1.5)*	***Communication(1.5)***	Confidence ***(1)***	Response to Questions(1)	***Total(10)***
Weightage	*25%*	25%	15%	15%	10%	*10%*	*100%*
Q1							
Q2							

Code of Conduct:

I S. V. K. Sharon Surya(192324246)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S. Sharon Surya

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

(Each question aligned with BL3–BL4, 0.75 Marks each)

Q1. Dataset: Retail Transactions

TID	**Items**
T1	Milk, Bread
T2	Milk, Butter
T3	Bread, Butter
T4	Milk, Bread, Butter
T5	Milk, Eggs

Question:
Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example. *(BL3 – 0.75 Marks)*

Q2. Dataset: Transaction Data

TID	**Items**
T1	A, B
T2	B, C
T3	A, C
T4	A, B, C
T5	B, C

Question:
Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used. *(BL4 – 0.75 Marks)*

Q3. Dataset: Transaction Data

TID	**Items**
T1	A, B
T2	A
T3	B

TID	Items
T4	A, B
T5	A

Question:
Explain support and confidence using the dataset and describe their importance in association rule mining. *(BL3 – 0.75 Marks)*

Q4 . Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question:
Explain multilevel association rules using the dataset and compare them with multidimensional rules. *(BL4 – 0.75 Marks)*

Q5. Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question:
Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues. *(BL4 – 0.75 Marks)*

Q1. Association Rule Mining in Retail

Association rule mining helps a retail store discover relationships among products that customers frequently purchase together. Measures such as support and confidence identify useful product combinations and rules.

In the given dataset, Milk occurs in T1, T2, T4 and T5, so its support is 4/5 = 80%. Bread occurs in T1, T3 and T4, so its support is 3/5 = 60%. The combination {Milk, Bread} occurs in T1 and T4, so its support is 2/5 = 40%.

A practical rule is:
Milk → Bread

Its confidence is:
Confidence(Milk → Bread) = Support(Milk, Bread) / Support(Milk)
= 2/4 = 50%.

A store could use such information for product placement, promotions, bundle offers, or recommendations. For example, if customers who buy Milk frequently also buy Bread, the store can place the products nearby or offer a combined promotion.

R PROGRAM

```
# Q1: Retail association rule measures
transactions <- list(
  c("Milk","Bread"),
  c("Milk","Butter"),
  c("Bread","Butter"),
  c("Milk","Bread","Butter"),
  c("Milk","Eggs")
)

support <- function(itemset)
  mean(sapply(transactions, function(t) all(itemset %in% t)))

milk_bread_support <- support(c("Milk","Bread"))
milk_support <- support(c("Milk"))
confidence <- milk_bread_support / milk_support

cat("Support(Milk) =", milk_support, "\n")
cat("Support(Milk, Bread) =", milk_bread_support, "\n")
cat("Confidence(Milk -> Bread) =", confidence, "\n")
```

Q2. Apriori vs FP-Growth

Apriori and FP-Growth are both frequent-itemset mining algorithms, but they use different approaches.

Apriori uses a level-wise process. It generates candidate itemsets and repeatedly scans the database to calculate support. It applies the Apriori property: if an itemset is infrequent, all of its supersets are also infrequent. It is simple and easy to understand, but candidate generation and repeated database scans can become expensive for large datasets.

FP-Growth avoids explicit candidate generation. It compresses transactions into an FP-Tree and mines the tree using conditional pattern bases and conditional FP-Trees. This generally reduces database scans and can be more efficient for large or dense transaction datasets.

For the given data, Apriori is suitable when the dataset is small and transparency of candidate generation is important. FP-Growth is preferable when the transaction database is large and candidate generation would create too many itemsets.

R PROGRAM

```
# Q2: Compare frequent patterns for the transaction data
transactions <- list(
  c("A","B"),
  c("B","C"),
  c("A","C"),
```

```
  c("A","B","C"),
  c("B","C")
)

support_count <- function(itemset)
  sum(sapply(transactions, function(t) all(itemset %in% t)))

items <- c("A","B","C")

cat("1-itemset support counts:\n")
for (x in items)
  cat("{",x,"} = ",support_count(x),"\n",sep="")

cat("\n2-itemset support counts:\n")
pairs <- combn(items, 2, simplify=FALSE)
for (x in pairs)
  cat("{",paste(x,collapse=", "),"} = ",
      support_count(x),"\n",sep="")

cat("\nFP-Growth concept: transactions are compressed into an FP-Tree,\n",
    "then conditional pattern bases are mined without candidate generation.\n")
```

Q3. Support and Confidence

Support measures how frequently an itemset occurs in the complete transaction database.

Support(X) = Number of transactions containing X / Total number of transactions.

Confidence measures how often the consequent occurs when the antecedent occurs.

Confidence(A → B) = Support(A ∪ B) / Support(A).

In the dataset, A occurs in T1, T2, T4 and T5, so:
Support(A) = 4/5 = 80%.

B occurs in T1, T3 and T4, so:
Support(B) = 3/5 = 60%.

Both A and B occur together in T1 and T4, so:
Support(A,B) = 2/5 = 40%.

Therefore:
Confidence(A → B) = (2/5)/(4/5) = 2/4 = 50%.

Support tells us how common a pattern is, while confidence tells us how reliable an association rule is. Both are important for removing weak or insignificant rules.

R PROGRAM

```
# Q3: Calculate support and confidence
transactions <- list(
  c("A","B"),
  c("A"),
  c("B"),
  c("A","B"),
  c("A")
)

support <- function(itemset)
  mean(sapply(transactions, function(t) all(itemset %in% t)))

support_A <- support(c("A"))
support_B <- support(c("B"))
support_AB <- support(c("A","B"))
confidence_A_B <- support_AB / support_A
```

```
cat("Support(A) =", support_A, "\n")
cat("Support(B) =", support_B, "\n")
cat("Support(A,B) =", support_AB, "\n")
cat("Confidence(A -> B) =", confidence_A_B, "\n")
```

Q4. Multilevel Association Rules vs Multidimensional Rules

Multilevel association rules discover relationships at different levels of a concept hierarchy. In the given dataset, Electronics is a general-level concept, while Mobile and Laptop are more specific concepts.

At the higher level:
Electronics appears in all five transactions, so its support is 5/5 = 100%.

At the lower level:
Mobile appears in T1, T3 and T4, so support(Mobile) = 3/5 = 60%.
Laptop appears in T2 and T5, so support(Laptop) = 2/5 = 40%.

Thus, a multilevel analysis can move from the general concept Electronics to specific products such as Mobile and Laptop. This helps discover both broad and detailed patterns.

Multidimensional association rules use multiple attributes or dimensions, such as Product, Customer Age, Location, and Time. Multilevel rules instead focus on different abstraction levels within a concept hierarchy. Therefore, multilevel mining is hierarchy-oriented, whereas multidimensional mining combines different attributes or dimensions.

R PROGRAM

```
# Q4: Multilevel support calculation
transactions <- list(
  c("Electronics","Mobile"),
  c("Electronics","Laptop"),
  c("Electronics","Mobile"),
  c("Electronics","Mobile"),
  c("Electronics","Laptop")
)

support <- function(itemset)
  mean(sapply(transactions, function(t) all(itemset %in% t)))

cat("Level 1 - Electronics support = ",
    support(c("Electronics")), "\n", sep="")

cat("Level 2 - Mobile support = ",
    support(c("Mobile")), "\n", sep="")

cat("Level 2 - Laptop support = ",
    support(c("Laptop")), "\n", sep="")

cat("\nExample hierarchy:\n")
cat("Electronics\n")
cat("  |-- Mobile\n")
cat("  |-- Laptop\n")
```

Q5. Challenges in Mining Large Datasets and Scalable Methods

Mining large datasets creates several challenges: very large numbers of transactions and items, high memory requirements, repeated database scans, a huge candidate search space, and increased execution time. Noise and redundant patterns can also make the final rule set difficult to interpret.

For the given sample, combinations of A, B and C must be examined. As the number of items grows, the number of possible itemsets increases rapidly. This causes the search space and computation cost to grow.

Scalable methods address these problems by reducing unnecessary candidate generation, compressing the transaction database, partitioning the data, using efficient data structures, and processing data in parallel or distributed environments. Apriori can use pruning to reduce candidates, while FP-Growth compresses transactions into an FP-Tree and avoids explicit candidate generation.

For very large datasets, scalable approaches such as FP-Growth, parallel/distributed mining, and database-oriented processing can reduce execution time and memory pressure.

R PROGRAM

```
# Q5: Demonstrate itemset support and search-space growth
transactions <- list(
  c("A","B","C"),
  c("A","B"),
  c("B","C"),
  c("A","C"),
  c("A","B","C")
)

support_count <- function(itemset)
  sum(sapply(transactions, function(t) all(itemset %in% t)))

items <- c("A","B","C")

cat("Support counts:\n")
for (k in 1:length(items)) {
  sets <- combn(items, k, simplify=FALSE)
  for (s in sets) {
    cat("{",paste(s,collapse=", "),"} = ",
        support_count(s),"\n",sep="")
  }
}

cat("\nNumber of possible non-empty itemsets for n items is 2^n - 1.\n")
n <- 3
cat("For n =", n, ", possible itemsets = ", 2^n - 1, "\n", sep="")

cat("\nScalability idea: use pruning, FP-Tree compression,\n",
    "partitioning, and parallel/distributed processing for large data.\n")
```

Submission note: The original uploaded Assessment Tool 3 document is preserved at the front. Each question is followed by its answer and the corresponding R program.